# Modelling Wildfire Propagation Using Multicriteria Evaluation and Cellular Automata: A Case Study of the 2003 Paradise Wildfire

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**Abstract:**
Wildfire frequency and intensity has increased globally, necessitating advanced modeling tools for prediction and management. Multicriteria Evaluation Cellular Automata(MCE-CA) models offer a promising approach to simulate wildfire propagation by integrating additional datasets and representing dynamic environmental factors. This study applies the MCE-CA model to the 2003 Paradise wildfire demonstrating its ability to capture key spatial dynamics, with an overall predictive accuracy of 88.98% and a Figure of Merit(FOM) of 0.8834. The results highlight strengths in capturing key spatial dynamics while also addressing limitations in predictive accuracy, underscoring the need for refined datasets and modeling techniques to improve real-world applicability.

## 1. Introduction

Devastating wildfires are increasing in frequency and intensity, necessitating advanced models to enhance wildfire management, prevention and mitigation efforts. Predicting wildfire propagation remains a significant challenge due to the interconnected and complex factors playing a role in its propagation, including topography, vegetation, and land development, all of which exhibit variation over space and time. Multicriteria Evaluation Cellular Automata (MCE-CA) models offer a promising approach by integrating spatial datasets and temporal dynamics to simulate fire behavior. Using the 2003 paradise wildfire as a case study, this research examines the potential of Multicriteria Evaluation(MCE) Cellular Automata (CA) models to represent and forecast wildfire propagation, focusing on their ability to capture the complexities of dynamic spatial phenomena influenced by environmental and temporal factors.

## 2. Literature review and theoretical background

### 2.1.Wildfire modeling Approaches

Simulating wildfire propagation requires models capable of capturing spatially and temporally dynamic processes. Cellular Automata(CA) has emerged as a solution for modelling wildfire spread due to its adaptability and ability to represent complex phenomena. CA models simplify complex systems into discrete cell-based grids governed by transition rules (Freire & DaCamara, 2019). However, despite the large application of CA models, it becomes evident that there are several challenges faced by the model representing the dynamic and interconnected nature of wildfire propagation (Ghosh et al., 2024).

### 2.2. Key theoretical concepts

CA models divide the landscape into a grid of cells, each governed by local transition rules that determine state changes over time, making it capable of modeling complex spatial phenomena. This structure supports localized interactions but often falls short of fully incorporating complex real-world dynamics. MCE addresses this limitation by integrating diverse spatial datasets, such as vegetation tupe and topographical features within its transition rules. By weighting and prioritizing multiple factors, MCE-CA provides an even more adaptive framework for simulating wildfire dynamics, allowing for a more nuanced understanding of how multiple factors influence fire spread (Freire & DaCamara, 2019; Ghosh et al., 2024) .

### 2.3 Existing research

Research in CA models has demonstrated its effectiveness in spatial modelling, specifically in more controlled contexts where environmental factors remain more static (Xu et al., 2022). However, when the environmental conditions become more dynamic the addition of MCE increases the number of scenarios in which CA can be meaningfully applied, as it allows for the integration of more datasets. Despite the integration of MCE, gaps remain, particularly in accounting for rapidly changing conditions, such as wind speed and moisture levels, or emergent behaviours like feedback loops. Addressing these gaps requires further exploration of hybrid approaches (Ghosh et al., 2024).

## 3. Datasets

### 3.1. Primary data

**Fire Progression Data:** 2 consecutive vector files showing the wildfire's progression over a 24 hour period, serving as a basis for temporal analysis *Figure 1.*

### 3.2. Auxiliary Data

**Digital Elevation Models(DEMs):** 9 DEM tiles were stitched together to create a seamless elevation surface accounting for the terrain's slope and aspect *Figure 2.1.*

**Land Use Map:** A land use map of California, *Figure 3*, classifying both the different types of vegetation and human development in the region.

### 3.3. Data Processing Techniques

**DEM Stitching:** A seamless elevation surface was created by merging multiple DEM tiles, shown in *figure 2.2.*

**Land Use Map Simplification:** The map was reclassified into broader categories to enhance suitability modeling and improve accuracy*(Subfigures 4.1,4.2,4.3)*.

**Geospatial Alignment:** To ensure consistency across datasets, geospatial alignment was performed, georeferencing all spatial layers to maintain proper alignment relative to the study area. The data sets were converted to raster format and standardized to a uniform resolution ensuring compatibility within the CA model framework.

## 4. Methodology

### 4.1. Study Site Descriptions

This study focuses on the 2003 Paradise wildfire, characterized as having diverse terrain, including varying elevations, regions of different degrees of development, and a variety of different land cover. These spatial and temporal characteristics create an ideal environment to explore how different factors including elevation, vegetation and human development may influence wildfire propagation.

### 4.2.Model Framework

The MCE-CA model forecasts the Paradise wildfire's progression over time by combining CA's ability to simulate spatial processes with MCE's consideration of multiple spatial factors. The CA framework models wildfire dynamics as cell transitions between discrete states (unburnt, burning and burnt) with each cell governed by transition rules that incorporate influences from neighboring cells. MCE enhances this framework by incorporating spatial factors (slope, vegetation and human development). By weighting these factors, MCE improves model accuracy by accounting for the influence of various environmental components on wildfire propagation.

### 4.3. Simulation Setup

1. **Temporal Resolution:** The simulation was set up with a 24-hour temporal resolution, based on the fire progression data.
2. **Suitability Map:** A suitability map was created using processed data from the DEM and Land Use maps, designed to incorporate slope, land development to assess their influence on wildfire propagation.
3. **Constraint Map**: A constraint map was created from the land use map identifying waterways, preventing the wildfire from unnaturally spreading across bodies of water.

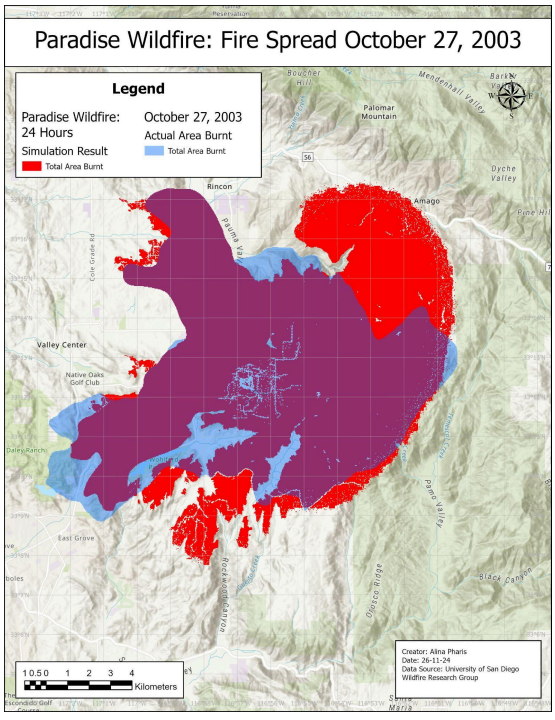
### 4.5. Model Transition Rules

1. **Stochastic Disturbance Rule:** A stochastic disturbance rule simulates randomness by giving unburned cells within 5-15 cells of a burning cell a 20% chance of igniting, simulating erratic wildfire behaviour like floating embers or falling tree branches.
2. **Neighbourhood Transition Rule:** A custom 13x13 neighborhood capturing fire spread based on wind direction and slope.was defined to simulate the spread of fire based on the wind direction and slope, visualized in. If at least one neighboring cell is burning, the central unburned cell has a 90% chance of igniting in the next iteration. *Figure 7*
3. **Cell-Level Transition Rule:** Once a cell is burning it has a 50% chance of transitioning to a burnt state, tracking the fire's progression.
4. **Allocation and Quantity Rule:** *Figure 5* The MCE framework allocates weights to factors influencing fire spread in a suitability map, weighting elevation as 75% favoring fire spread uphill(*Subfigure 5.2)*, and urban infrastructure at 25% to resist fire spread*(Subfigure 5.3)*. Cells must have a minimum suitability of 0.6 to burn.
5. **Constraint Rule:**A binary map prevents fire from spreading over waterways ensuring the model remains geographically realistic, avoiding unnatural spread across water bodies *Figure 6*.

## 5. Simulation Results

### 5.1. Key Findings

The simulation results, shown in *Figure 8,* demonstrate the progression of wildfire spread over a series of consecutive time intervals. The simulation captures the dynamics of fire behaviour, providing visual representation of changes across the landscape over time. The comparison of the simulated fire spread with real-world data indicates that the model closely approximates the general pattern of the fire's progression.




*Final Simulation Results compared to Actual Fire Spread*

### 5.2. Strengths

The MCE model exhibited several strengths:

- **High Overall Accuracy:** Achieving an accuracy of 0.8898, the model demonstrated a strong alignment between the simulated and real-world data.
- **Integration of Multiple Datasets:** By incorporating DEMs and land use data, the model successfully accounted for terrain and vegetation enhancing predictive reliability
- **Geospatial Consistency:** The alignment and preprocessing of spatial datasets ensured the model maintained a consistent geospatial relationship across all layers.

### 5.3. Limitations

Despite its strengths, the model faced several limitations:

- **Sensitivity to Input Quality:** The model's accuracy was heavily dependent on the quality of input datasets, particularly the land use data.
- **Simplified Reclassification:** The generalization of land use categories may have introduced inaccuracies in representing finer-scale variations.
- **Predictive Challenges:**While the model performed well overall, certain areas showed discrepancies between simulated and real world outcomes suggesting that the model did not fully capture the complexity of wildfire behavior.

## 6. Discussion

### 6.1. Interpretation of results

The simulation results of the MCE-CA model show both promising capability and critical areas for improvement in wildfire propagation modeling. The model's overall accuracy of 0.8898 reflects a strong agreement between simulated and real-world data, showing a successful integration of spatial datasets accounting for factors influencing wildfire propagation (slope, land development, and land cover). The Figure of Merit(FOM = 0.8834) supports a strong alignment with real-world data, especially in regions of persistence. Despite this success, areas of misalignment point to challenges in capturing detailed fire behavior, suggesting that some factors such as vegetation or weather were over simplified or inadequately represented in the input data. These findings highlight the need to refine certain aspects of the model, however, with further adjustments, the model has the potential to become an effective tool for wildfire propagation modelling

### 6.2. Practical Implications

The findings emphasize the potential of CA-based models as tools for wildfire management and planning. High accuracy and spatial consistency make these models suitable for predicting wildfire spread and identifying high-risk areas. The results underscore the importance of integrating diverse datasets, such as terrain and vegetation maps, to enhance model reliability. Furthermore, the model's strengths in simulating persistence can aid in planning containment strategies and resource allocation.

### 6.3. Next Steps

Building on the successes of the study, there are several promising directions for continued development and enhancement of the MCE-CA model :

**Incorporate Real-Time Data**: Integrate real-time weather variables, such as wind speed and humidity, to refine the model's responsiveness and accuracy.

**Refine Classifications and Model Parameter:** Future research will focus on revisiting the land use reclassification methods to finer scale details that may have been lost in thor study, as well as adjusting the parameters to account for more complex nonlinear interactions between environmental factors.

**Expand Validation:** Test the model across different geographic regions and fire scenarios to evaluate its generalizability and robustness.

**Explore Hybrid modeling approaches:** Finally we will investigate the potential of combining CA models with machine techniques, such as neural networks, or machine learning technologies, to capture nonlinear relationships and spatial complexities that are more challenging to model using traditional CA frameworks alone.

By taking these next steps, we aim to refine our CA-based model to become a more powerful tool for wildfire management, providing decision makers with enhanced predictive capabilities and better preparation for future wildfire events.

## 7. Conclusion

### 7.1. Summary

The application of Cellular Automata (CA) modeling to simulate wildfire progression demonstrated significant promise in capturing large-scale patterns of fire spread. With an overall accuracy of 0.8898 and a Figure of Merit (FOM) of 0.8834, the model provided reliable predictions of both fire persistence and progression. However, limitations such as data sensitivity, simplification of key variables, and occasional misalignment between simulated and real-world outcomes highlight areas for improvement. These findings reinforce the utility of CA-based models while emphasizing the need for ongoing refinement to enhance predictive accuracy and adaptability.

## 8. Acknowledgements

This study would not have been possible without the valuable datasets provided by: U.S. Department of the Interior's Burned Area Emergency Stabilization and Rehabilitation (BAER) team for the wildfire GIS datasets, made available through San Diego State University's GIS Data Center for Wildfire Recovery and Management.

A special thanks to Dr. Dragicevic and Alysha van Duynhoven for their guidance and support through the research process.

## 9. References

Freire, J. G., & DaCamara, C. C. (2019). Using cellular automata to simulate wildfire propagation and to assist in fire management. Natural Hazards and Earth System Sciences, 19(1), 169–179. https://doi.org/10.5194/nhess-19-169-2019

Ghosh, R., Adhikary, J., & Chemlal, R. (2024). Fire spread modeling using probabilistic cellular automata. In Communications in computer and information science (pp. 45–55). https://doi.org/10.1007/978-3-031-56943-2_4

San Diego State University. (n.d.). GIS data download: San Diego wildfire recovery and management. https://map.sdsu.edu/firenet/files/datadownload.htm
Xu, L., Liu, X., Tong, D., Liu, Z., Yin, L., & Zheng, W. (2022). Forecasting urban land use change based on cellular automata and the PLUS model. Land, 11(5), 652. https://doi.org/10.3390/land11050652
Péré, C., et al. (2020). Predicting wildfire behavior using cellular automata-based models: A review. Forests, 11(9), 986.

**10. Figures**

*Figure 1: Paradise Wildfire Original Fire Maps*

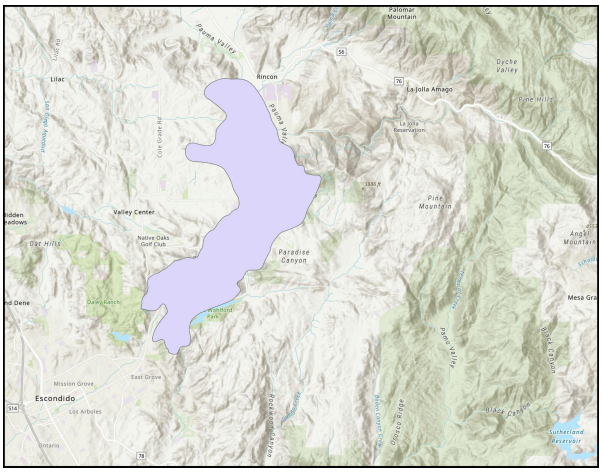



*Figure 1.1: October 26, 2003*

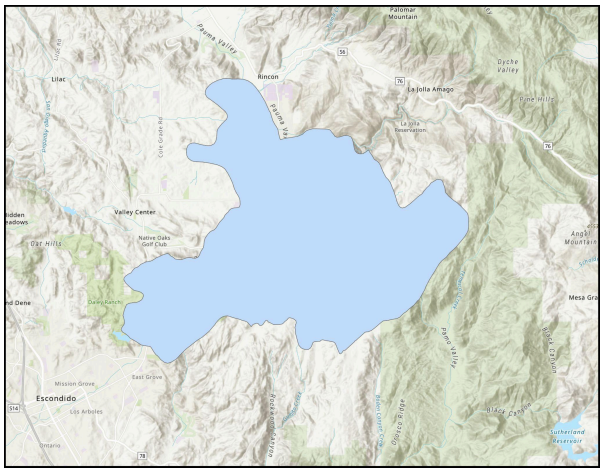



*Figure 1.2: October 27, 2003*

*Figure 2: Digital Elevation Models*

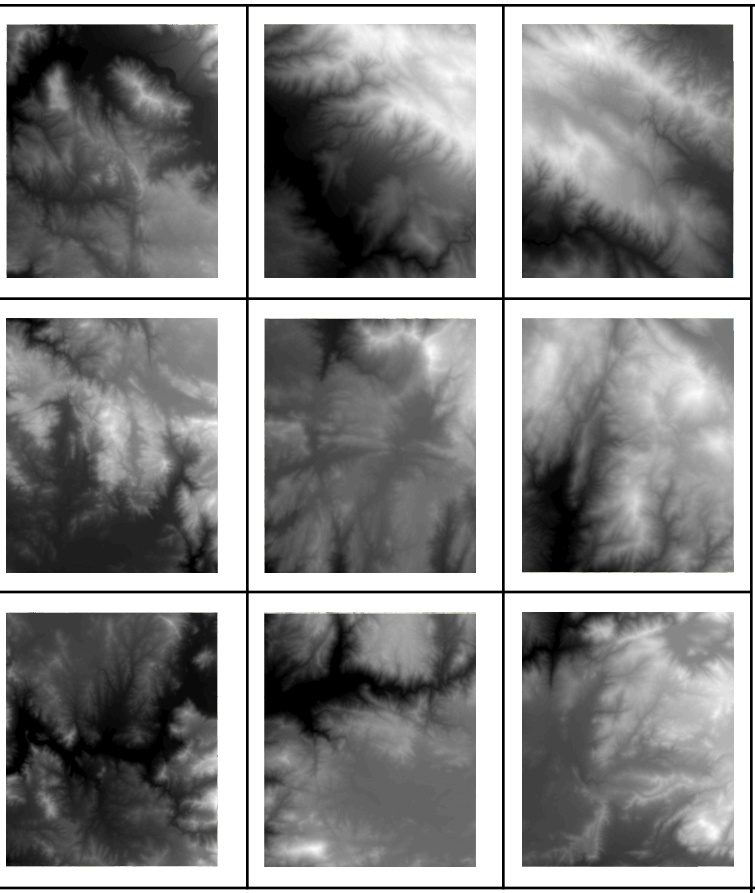


*Figure 2.1: Individual DEMs*

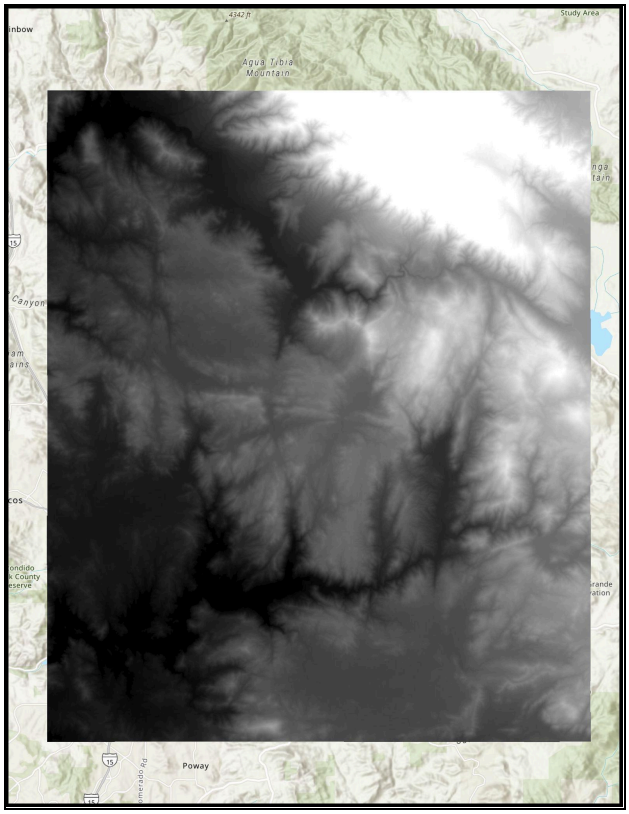



***Figure 2.2**: Stitched DEM of Study Area*

*Figure 3: Original California Land Use Map*



*Figure 4: Simplified Land Use Maps*

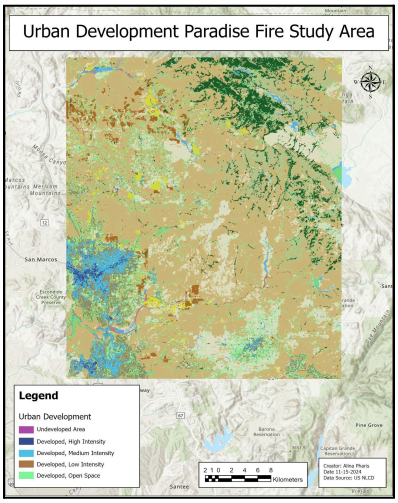



*Figure 4.1: Urban Development Map*

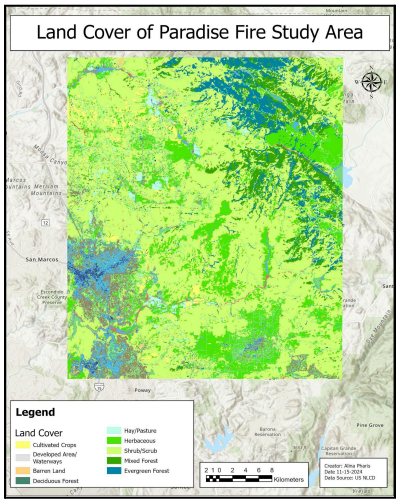



*Figure 4.2: Land Cover Map*

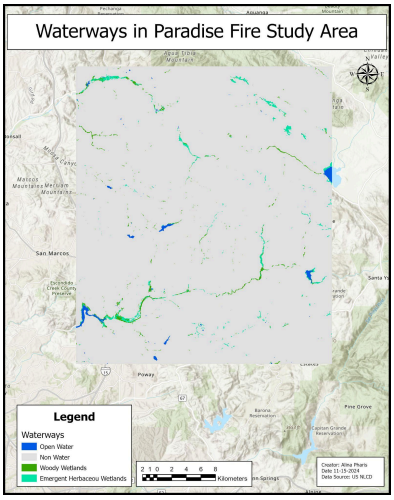



*Figure 4.3: Waterways Map*

*Figure 5: Suitability Maps*

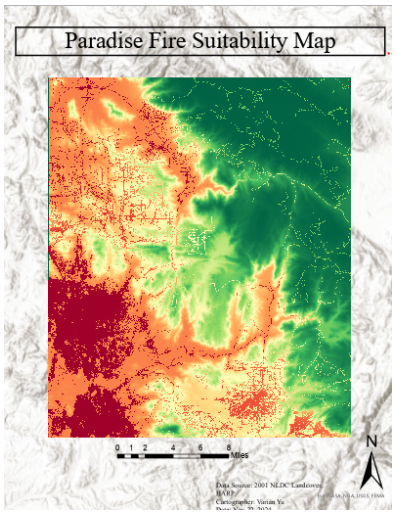



*Figure 5.1: Complete Suitability Map*

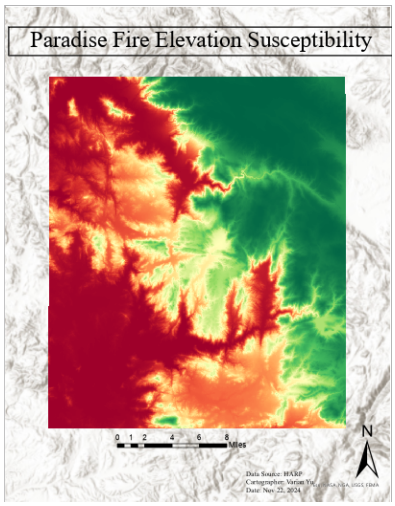



*Figure 5.2: Elevation Suitability Map*




*Figure 5.3: Urban Infrastructure Suitability Map*

*Figure 6.1: Waterways Constraint Map*

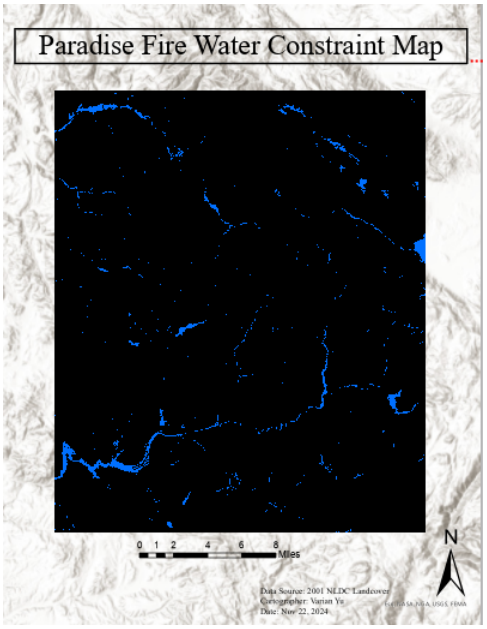
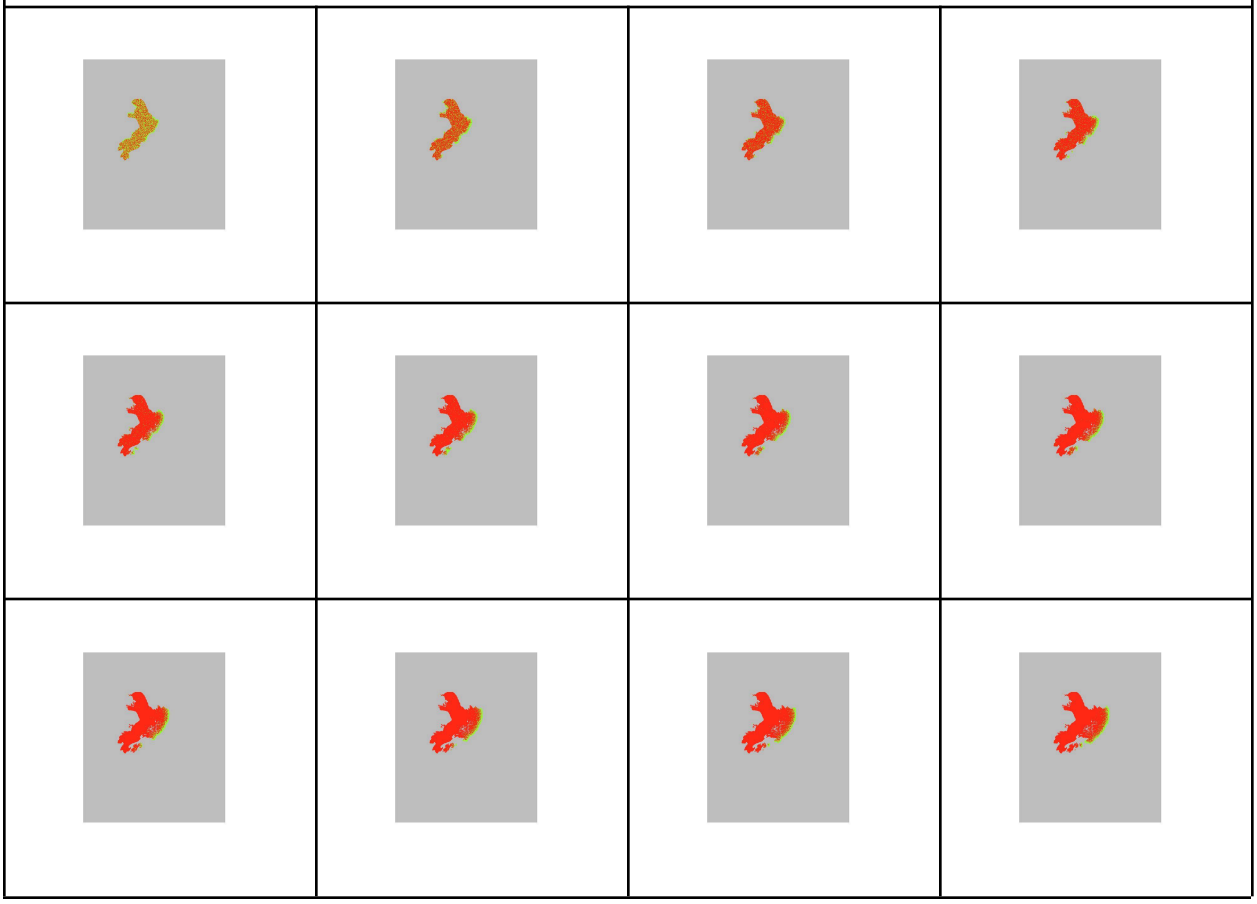



*Figure 7: Custom Neighborhood Definition*

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |

*Figure 8: 24 Iterations of Simulated Fire Progression*



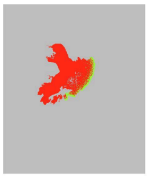
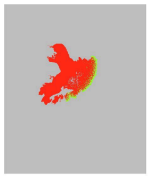
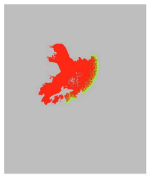
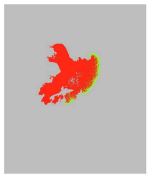
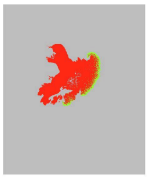
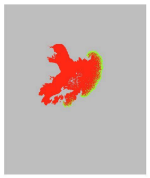
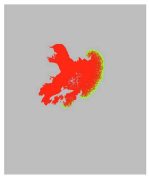
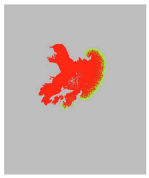
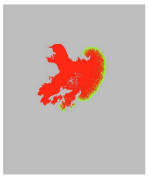
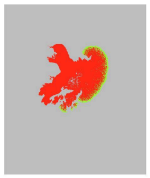
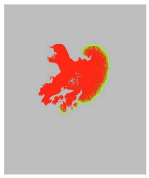
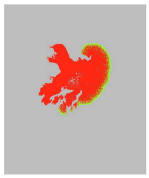
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|  |  |  |  |
|  |  |  |  |

*Figure 9: Model Evaluation Accuracy Assessment*

## Two-Map Comparison Results

### Accuracy Assessment

| Name | Metric Value |
|---|---|
| Overall Accuracy | 0.8898 |
| Number of Agreeing Cells | 1442574 |
| Number of Disagreeing Cells | 178610 |
| Number of Excluded Cells | 0 |

### Class-Level Accuracy Measures

| Name | Metric Value |
|---|---|
| Class 0 | 1 |
| Class 2 | 0.0946 |
| Class 3 | NaN |

### Kappa Measures

| Name | Metric Value |
|---|---|
| Kappa | 0.515 |
| Kappa Histogram | 0.515 |
| Kappa Location | 1 |

### Cross Tabulation

| | Simulated Class 0 | Simulated Class 2 | Simulated Class 3 |
|---|---|---|---|
| **Real Class 0** | 1423913 | 0 | 0 |
| **Real Class 2** | 0 | 18661 | 178610 |
| **Real Class 3** | 0 | 0 | 0 |

*Figure 9.1: 2 Map Comparison*

## Three-Map Comparison Results

### Change Accuracy Assessment

| Name | Metric Value |
|---|---|
| Overall Accuracy | 0.8898 |
| Change Forecasting Accuracy | 0.8834 |
| Persistence Forecasting Accuracy | 1 |

### Change Error Assessment

| Name | Metric Value |
|---|---|
| Total Number of Cells Compared | 1621184 |
| Hits (H) | 1353109 |
| Misses (M) | 0 |
| Incorrect Changes (IC) | 178610 |
| False Alarms (FA) | 0 |
| Correct Rejections (CR) | 89465 |
| Total Error (TE) | 0 |
| Error due to Quantity (EQ) | 0 |
| Error due to Allocation (EA) | 0 |
| Excluded Cells | 0 |

### Figure of Merit

| Name | Metric Value |
|---|---|
| Figure of Merit (FOM) | 0.8834 |
| Producer's Accuracy (PA) | 0.8834 |
| User's Accuracy (UA) | 0.8834 |

*Figure 9.2: 3 Map Comparison*